NAGIZAAZ SHAIK

GENERATIVE AI ENGINEER · FINTECH · LLM & RAG SYSTEMS · CREDIT RISK · AWS ML

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Currently on CPT (Curricular Practical Training) — authorized to work in the U.S.

Data engineer turned ML and GenAI engineer for regulated fintech: three years of data engineering at a financial-services client, then two years building ML and GenAI systems at credit unions. Shipped a governed LLM gateway, compliance RAG that cut research from 2 hours to under 20 minutes, and an ML platform that cut deployment failures 85%, plus AML, real-time payment and model-risk work. Doctoral researcher on how AI fails in regulated settings; 20 merged PRs in 18 open-source projects.

EXPERIENCE

ML Platform Engineer · Contract — *Jul 2025 – Jun 2026*

Wright-Patt Credit Union ($9B assets, 520K+ members), Fairborn, OH

- Built the credit union's internal GenAI gateway (Claude, OpenAI, Ollama) with evaluation guardrails, cost controls, and RAGAS grounding checks: **one governed path** for the 12–15-person IT team to ship GenAI features.
- Founded the AWS ML platform: SageMaker endpoints and Pipelines, MLflow registry with gated promotion and automated rollback, per-environment VPC isolation, all codified in Terraform with GitOps on ArgoCD; **deployment failures down 85%** in 60 days.
- Owned ML data security (AES-256 at rest and in transit, least-privilege IAM, audit logging); passed **two consecutive NCUA audits with zero findings**.
- Produced model-risk documentation in the style of SR 11-7 for ML systems: model inventory (MLflow registry), intended use, validation results, known limitations, approvals and rollback records, kept on file for NCUA examination.
- Mentored two engineers into ML and platform ownership through weekly architecture and code reviews.

Stack: Claude, OpenAI, Ollama, RAGAS, AWS SageMaker, MLflow, Terraform, ArgoCD, Python, FastAPI

Machine Learning Engineer · Contract — *Jul 2024 – Jun 2025*

Day Air Credit Union ($830M assets, 53K+ members), Kettering, OH

- Built a RAG pipeline over 500+ internal policy, NCUA and procedure documents (LangChain, vector search, Claude), used by the ~50-person staff; research went **from 2 hours to under 20 minutes**, ~400 questions per month.
- Built the first-pass consumer-loan underwriting model (XGBoost on SageMaker) as the first dedicated ML engineer, scoring **~1,500 applications per month** at **0.84 AUC** and flagging delinquent and pending payments for collections.
- Trained and supported XGBoost models for BSA/AML alert scoring (Verafin, **~3,000 alerts per month**), KYC onboarding risk, and real-time payment decisions over RTP and FedNow (via Corporate One); fused text and tabular signals and tuned cost-sensitive thresholds, with flagged transactions feeding SAR/CTR reporting.
- Set up SageMaker Model Monitor drift detection with custom baselines; it caught **two data-drift events** before they reached credit decisions.
- Automated weekly retraining in Airflow: rolling-window refit, holdout evaluation, and AUC / precision / calibration gates before each release.
- Benchmarked BERT, RoBERTa and DistilBERT, then fine-tuned the best for 14-class ticket intent, triaging **~2,000 member tickets per month** with misrouting down about 35%; FastAPI at **under 200 ms p99**.

Stack: Claude, LangChain, RAG, BERT, FastAPI, AWS SageMaker, XGBoost, Apache Airflow

Senior Data Engineer · promoted from Associate in two years — *Jul 2020 – Jun 2023*

Cognizant (financial-services client), Hyderabad, India

- **Cut average query runtime 60%** by leading the Oracle 11g to Amazon Redshift migration for five business units: a columnar schema with tuned distribution and sort keys, and 20+ stored procedures rebuilt as Airflow DAGs.
- Built 15 production ETL/ELT pipelines across Oracle, SQL Server, flat files and REST APIs, with data cleaning and standardization and Apache Kafka streaming ingestion; a multi-stage validation layer **caught a silent upstream corruption that had invalidated three months of financial reports**.
- Built incremental and change-data-capture loads scheduled in Airflow with retries and failure alerts, backed by data-quality checks (row counts, null and duplicate checks, source reconciliation) and unit and regression tests.
- Tuned slow SQL and Redshift queries (sort and distribution keys, rewrites, vacuum and analyze) and modeled star-schema reporting tables for finance and risk reporting.
- Handled the client's production data errors (root-caused failed loads, fixed data defects, wrote runbooks) and led code reviews for three engineers as primary client contact on two accounts.

Stack: Amazon Redshift, Apache Airflow, Apache Kafka, Oracle PL/SQL, SQL Server, Python

RESEARCH

Doctoral Researcher, Applied AI

Aug 2026 – Present

Belhaven University, Jackson, MS

– Research how LLM agents fail under adversarial pressure: what data and tools an agent can reach, whether its guardrails and permission boundaries hold, and what a forensic postmortem shows after a failure. Demonstrated that a planted label drives a LoRA-tuned financial classifier to the attacker's label on **92%** of items (five markets, three seeds, controls), while a standard confidence gate caught only about half the flips (working paper *Fragile in Different Ways*).

– Download and run **9 open-weight model families** locally (DeepSeek, GLM, Kimi, Nemotron, Qwen) to probe their dual-use risk, measuring where safety guardrails hold and where they can be turned to misuse. Found a sandbox bypass in the code executor of a widely used AI-agent framework, reported it via a GitHub security advisory, and traced the copied code into **25 open-source projects**; released guard-gap-detector (working paper *When Vendoring Defeats the Patch*).

PROJECTS

LLM Gateway | Python, FastAPI, Redis, Docker — github.com/shaikn6/llm-gateway

One OpenAI-compatible API over Anthropic, OpenAI, and Ollama with per-tenant caching, rate limiting, A/B routing, cost tracking, and audit logs; **428 tests at 99% line coverage**, including contract tests against the real provider SDK signatures.

Finance Agent Crew | Python, asyncio, Claude, FastAPI — github.com/shaikn6/finance-agent-crew

A ticker goes in, a sourced equity-research brief comes out: concurrent SEC EDGAR, market, and news collection feeds parallel fundamental, sentiment, and risk agents; a writer agent returns a typed BUY / HOLD / SELL report, and a failed source is recorded, not fatal. **333 tests at 99% coverage**, no real-LLM calls.

credit-arena: Credit-Default Models | Python, scikit-learn, paired bootstrap — github.com/shaikn6/credit-arena

Six model families on 30,000 card accounts (22% default), scored on AUC, calibration, business cost and fair lending. Gradient boosting beat logistic regression by **+0.0275 AUC** (95% CI 0.0187–0.0369) and cut expected cost 8%; stacking and segment specialists added no significant gain. Approval ratios of 0.89–0.96 against the 0.80 four-fifths rule.

fin-lora: Financial Text Classification | PyTorch, PEFT, Qwen2.5, FinBERT — github.com/shaikn6/fin-lora

A LoRA-tuned Qwen2.5-0.5B trained on tweets scored only 59.0% on news; retrained on news, tweets, and headlines, it reached **84.7%** vs 71.7% for TF-IDF and 73.6% for FinBERT (267-example news set, about ±5 points). A confidence cascade that sends only uncertain items to the LLM cut average latency **63%** for 0.4 points of accuracy. Live demo: hf.co/spaces/9mark9/fin-lora-attack-demo.

Open Source | 20 merged PRs across 18 projects, incl. crewAI (60K stars; a fix for a bug I reported), Ray (44K), DSPy (39K), MLflow (28K)

SKILLS

Generative AI: Claude, OpenAI, Ollama, Hugging Face Transformers, LoRA / PEFT fine-tuning, Qwen2.5, BERT, RoBERTa, DistilBERT, FinBERT, prompt engineering

RAG & LLM Systems: LangChain, embeddings, BM25 + dense hybrid search, reranking, HyDE, RAGAS, LLM gateways (caching, rate limits, cost tracking), guardrails

LLM Safety & Evaluation: red-teaming, jailbreak and prompt-injection testing, adversarial robustness, LLM-as-judge evals, guardrail and agent-sandbox analysis, model forensics

Agents: Multi-agent pipelines (asyncio, Claude), CrewAI contributor

Credit Risk & ML: XGBoost, scikit-learn, PyTorch, text + tabular fusion / ensembling, calibration, cost-sensitive thresholds, paired-bootstrap model comparison, fair-lending analysis, drift monitoring

Data Engineering: ETL/ELT, data cleaning and validation, Apache Kafka, Apache Airflow, Amazon Redshift, Oracle PL/SQL, SQL Server

AWS & MLOps: SageMaker (endpoints, Pipelines, Model Monitor), Redshift, S3, Glue, EKS, IAM, VPC isolation, MLflow, Airflow, Terraform, ArgoCD, Docker, FastAPI, GitHub Actions

Security & Compliance: BSA/AML monitoring (Verafin), instant payments (RTP, FedNow via Corporate One), model governance (gated promotion, automated rollback), AES-256 encryption, least-privilege IAM, audit logging, NCUA examinations

Languages: Python, SQL

EDUCATION & CERTIFICATIONS

Belhaven University — Doctor of Business Administration (DBA), Applied AI · GPA 4.0 / 4.0 — *Aug 2026 – Present*

University of Dayton — M.S. Computer Science · GPA 3.93 / 4.0 — *Aug 2023 – May 2025*

CMR Engineering College — B.Tech. Computer Science and Engineering · GPA 8.9 / 10 — *Aug 2016 – May 2020*

AWS Certified Machine Learning – Specialty • AWS Certified Data Engineer – Associate
GCP Professional Data Engineer • Snowflake SnowPro Advanced: Data Engineer • dbt Analytics Engineering